



FinInnovate

INNOVATION • TRANSPARENCY
DATA-DRIVEN

ROUND 02

CASE STUDY

Presentation by **Sherpa**

EXECUTIVE SUMMARY

Situation

FinInnovate is a Vietnam-based FinTech company that leverages expertise in data science, AI, and user-centered design to disrupt the traditional lending market. Its mission is to provide accessible, fair, and personalized credit services to individuals who are underserved by conventional banks.

The Problem

Our predictive models, trained on 50,000 applicant records, confirm that this segment carries a high default risk (40.5%) under traditional lending models. Our deep-dive analysis reveals a critical insight: for this group, it's not the size of their income that predicts risk, but how they manage their limited resources. Their creditworthiness is hidden in their discipline and financial habits, not their payslip.

Our Solution

FinConnect

Smart Group Lending

We propose FinConnect, a revolutionary micro-lending platform built on the principles of the proven Grameen model. Instead of relying on traditional credit scores, FinConnect leverages:



FinInnovate
INNOVATION · TRANSPARENCY
DATA-DRIVEN

Group Responsibility

Users form small groups, turning social trust into collateral.

Smart Matching

Our AI algorithm connects users into reliable groups based on location, trust scores, and social network analysis.

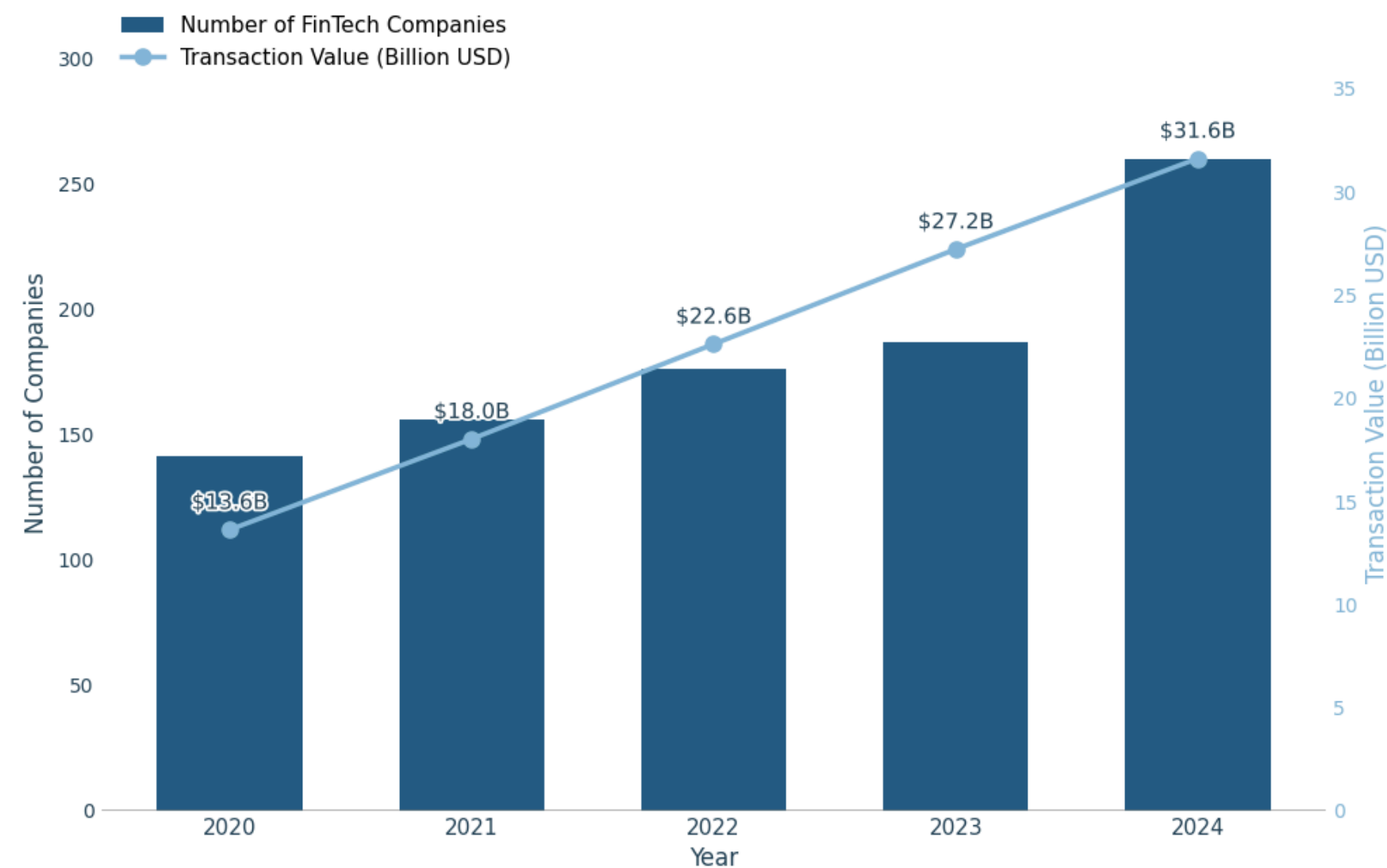
Dynamic Trust Scoring

We build a new kind of credit history based on user behavior and successful group loan cycles.

Vietnam's FinTech market is booming, becoming an attractive destination for international investors.

FinTech market is showing strong growth in both transaction volume and value.

Figure 1. Comprehensive Growth of FinTech in Viet Nam (2020-2024)



Data from Statista reveals an impressive increase in the number of FinTech companies, growing from 141 in 2020 to 260 in 2024—a **surge of over 84%**. Simultaneously, according to BDA Partners, the total FinTech transaction value has **more than doubled**, soaring from 13.6 billion USD in 2020 to **31.6 billion USD** in 2024.

The volume of cashless transactions reached a record level.

FIRST HALF OF 2025

12 BILLION

COMPARED TO THE SAME PERIOD IN 2024

7.83 BILLION ▲ 53.3%

Landmark policies came into effect on...

01/07/2025

Decree No. 94/2025/ND-CP

To officially launch a regulatory Sandbox for FinTech solutions in the banking sector.

Law on Value-Added Tax (2024)

All transactions, regardless of their value, must be made via cashless payment to be eligible for input VAT deduction.

Vietnam's FinTech market is booming, becoming an attractive destination for international investors.

For traditional banks, trust and scale are both a strength and a weakness.



- **Trust & Stability:** State-backed with a trusted history.
- **Physical Network:** Extensive branch network for secure, complex transactions.
- **Ecosystem:** All-in-one financial services.



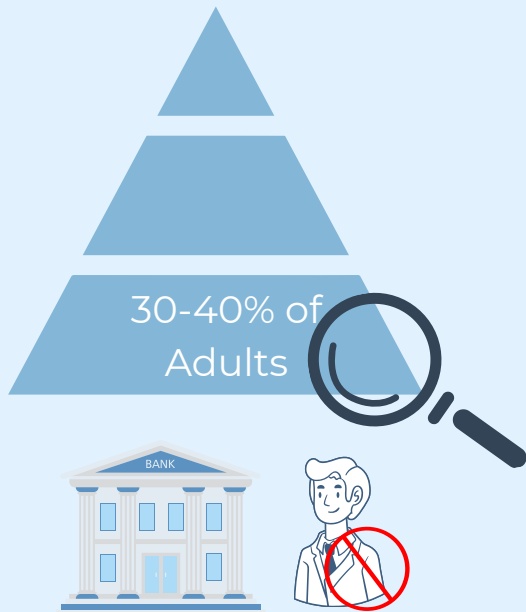
- **Slow & Inflexible:** Rigid, paper-based processes; limited hours.
- **High Costs:** Expensive infrastructure leads to higher fees.
- **Risk-Averse:** Rigid credit models exclude customers with no credit history.

Fintech's rapid growth is driven by solving the problems traditional banks create.



- ▶ Super-apps are building all-in-one platforms.
- ▶ Rise of focused players in P2P Lending, Insurtech, Wealthtech.
- ▶ Success hinges on turning data into personalized products.

The Strategic Opening



30-40% of Vietnamese adults lack a formal credit history. They need **small, flexible** loans but are often forced into predatory lending.

Banks find them too risky, and mainstream super-apps haven't yet targeted their specific needs.

The database we use includes 50,000 observations with 13 variables for randomly collected loans in the market.

Figure 1. Customers structure by target variable

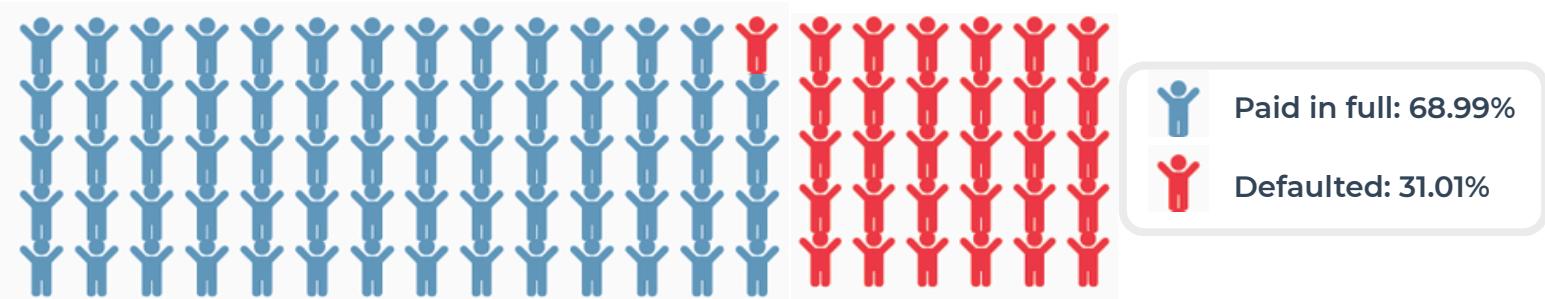


Figure 2. Customer current job tenure

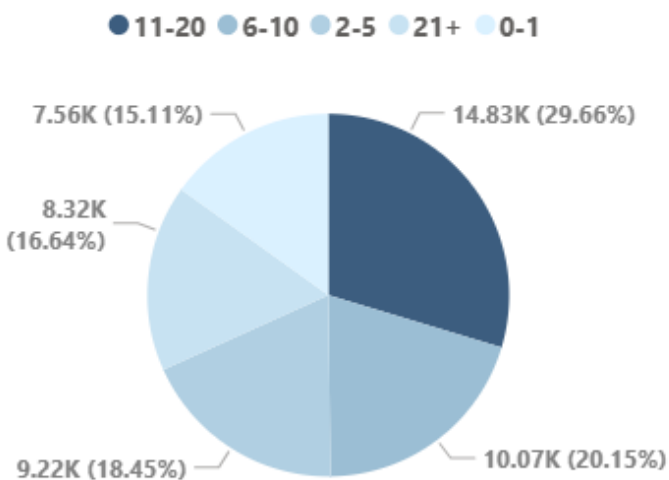


Figure 3. Customer home ownership status

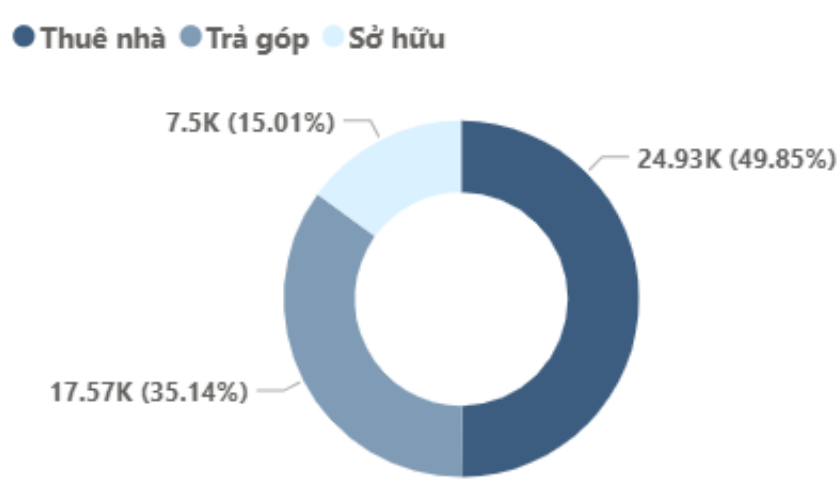


Figure 5. Customer credit rating

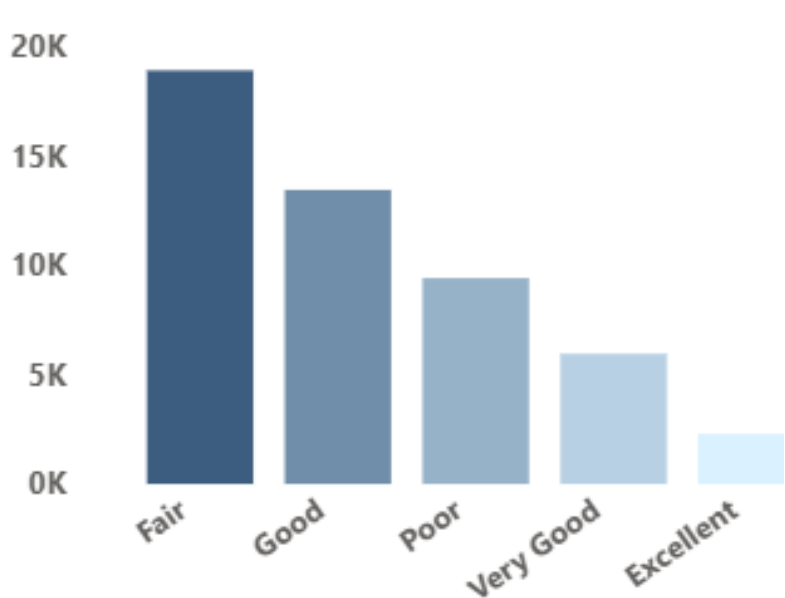
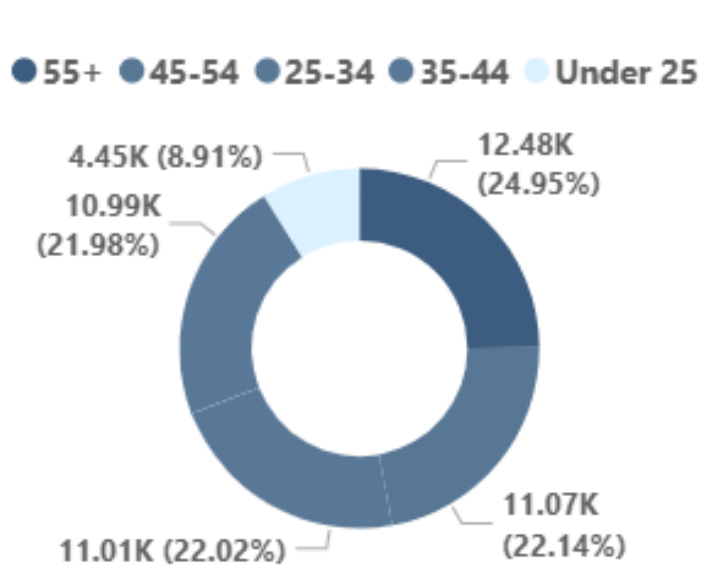


Figure 6. Customer age group

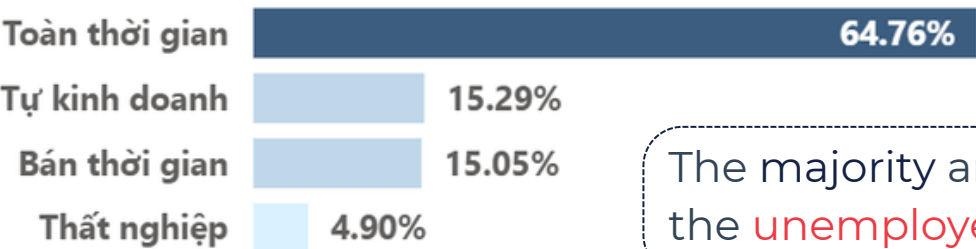


The portfolio is heavily skewed towards **medium-to-high risk (63%)**, with the Fair dominates the portfolio, carrying elevated risk.

The customer base is remarkably evenly distributed across four **adult** age groups.

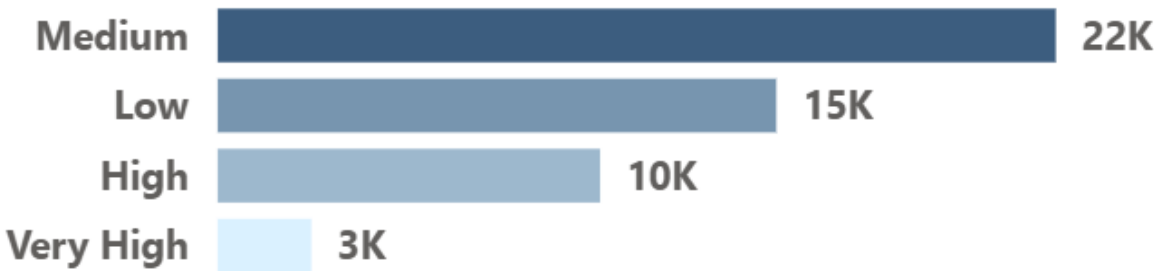
The majority (**~70%**) have **≤10** years of work experience (relatively young and early-career borrowers), a half rent house (**~50%**)

Figure 4. Customer employment status



The majority are **full-time** workers (**65%**); the **unemployed** form the smallest group (**5%**)

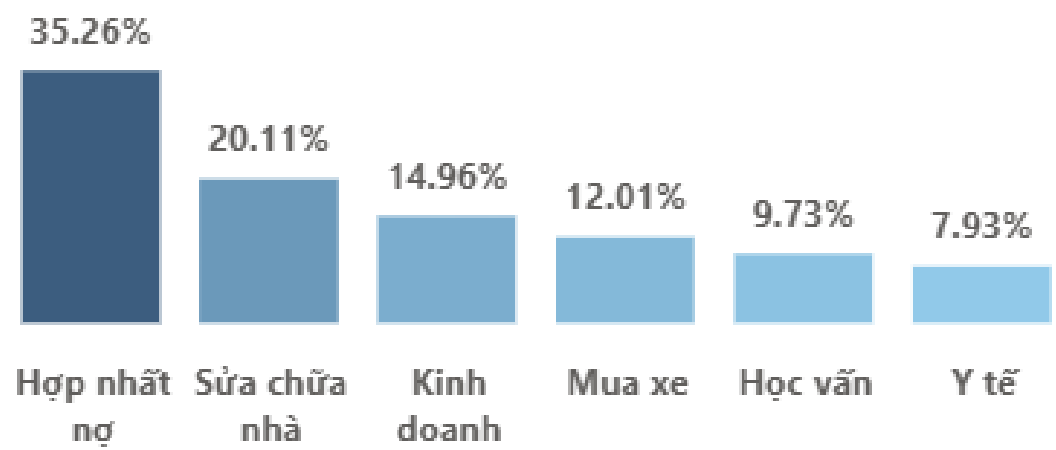
Figure 7. Customer income level



Customer base is concentrated in **medium** and **low** income, while **high** and **very high** remain small.

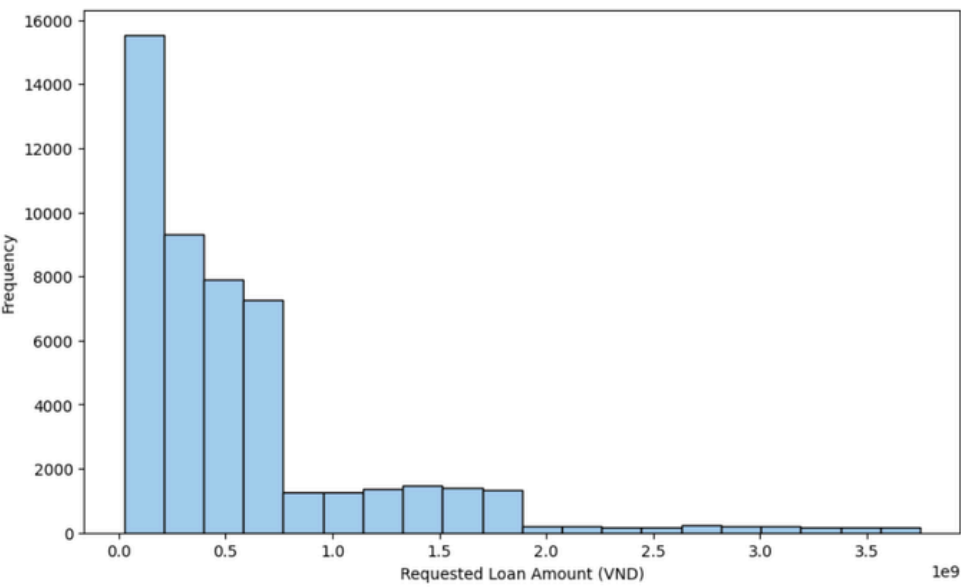
The dataset shows a core customer base of medium-income, full-time employees, renters, with Fair–Good credit scores.

Figure 8. Customer loan purpose



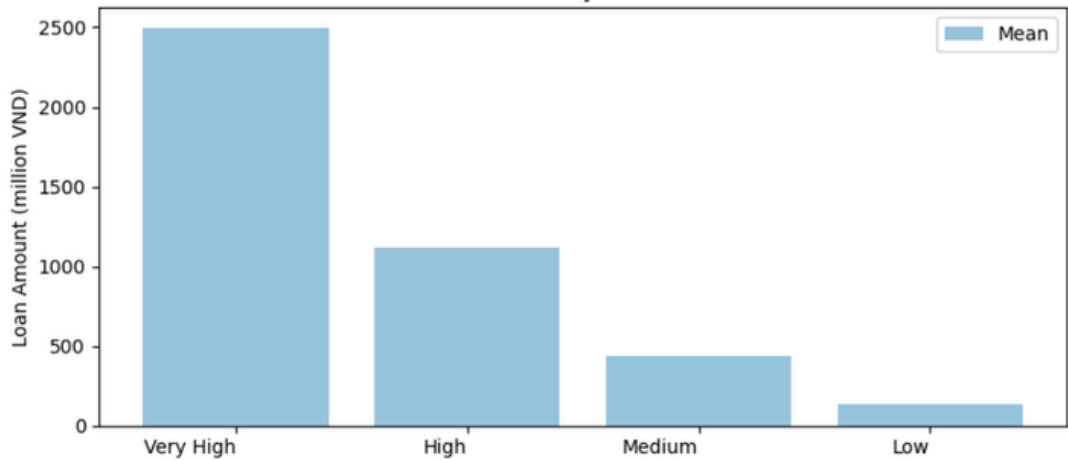
Most borrowers take loans mainly for debt consolidation

Figure 9. Customer requested loan



Most borrowers request small loans (< 500M VND), while large loans (>2B VND) are rare. Small and medium-sized loans dominate the market.

Figure 10. Customer requested loan by income level



Borrowers with higher income levels request while low-income ones with smaller loans. Loan amount strongly **correlates with** income level.

Figure 11. Customer default status by income

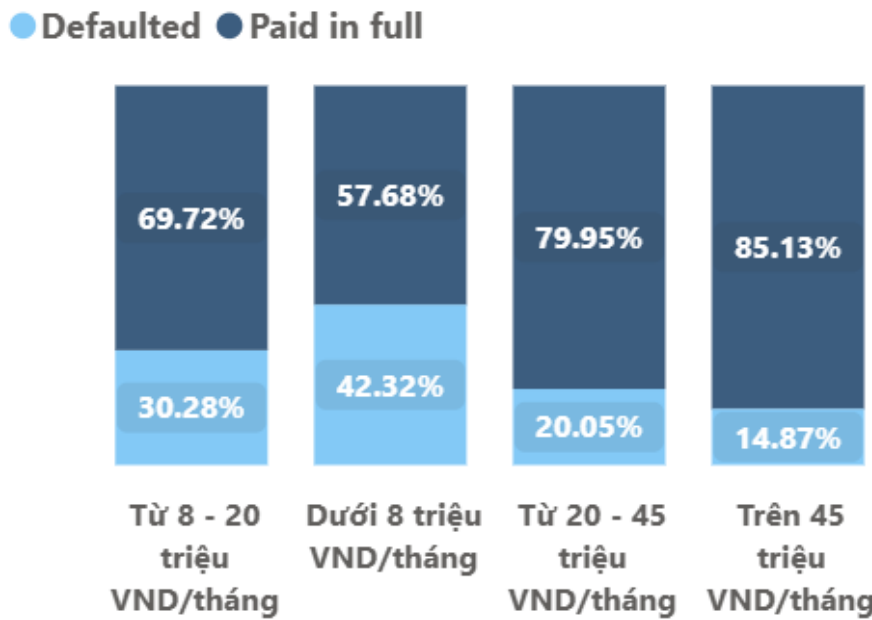


Figure 12. Customer default status by credit rating

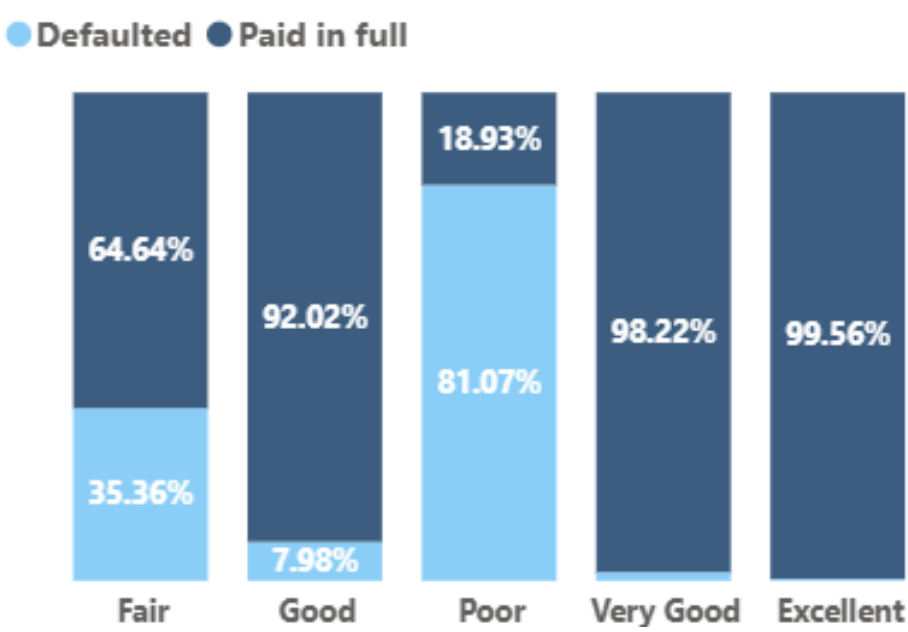


Figure 12. Customer default status by employment status

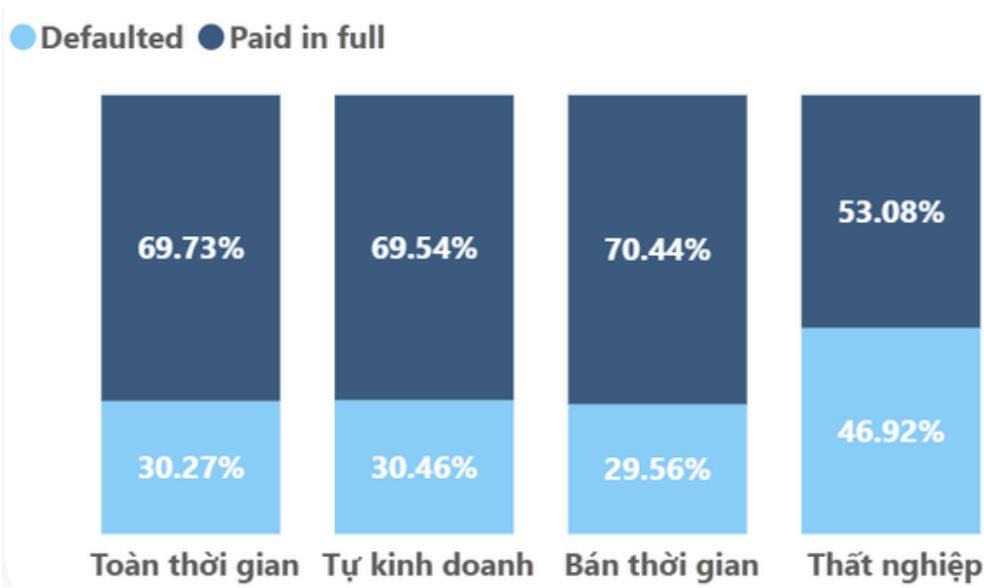
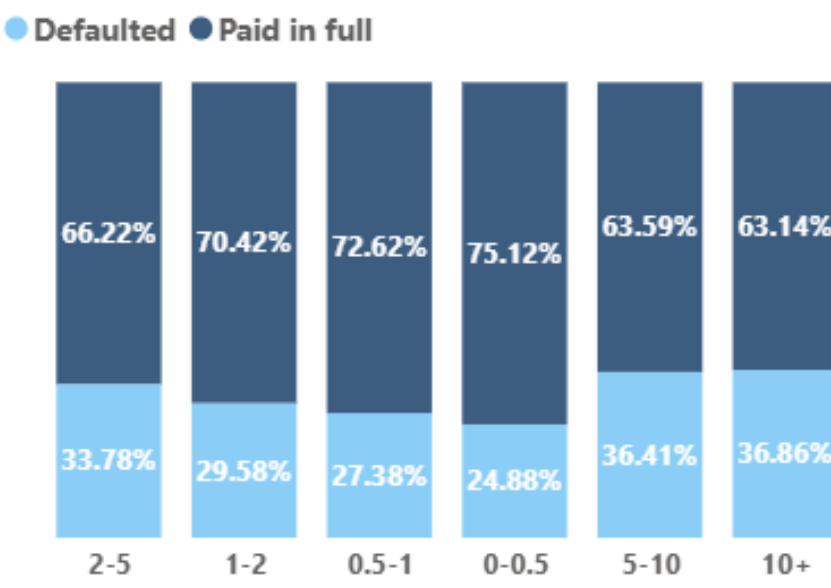
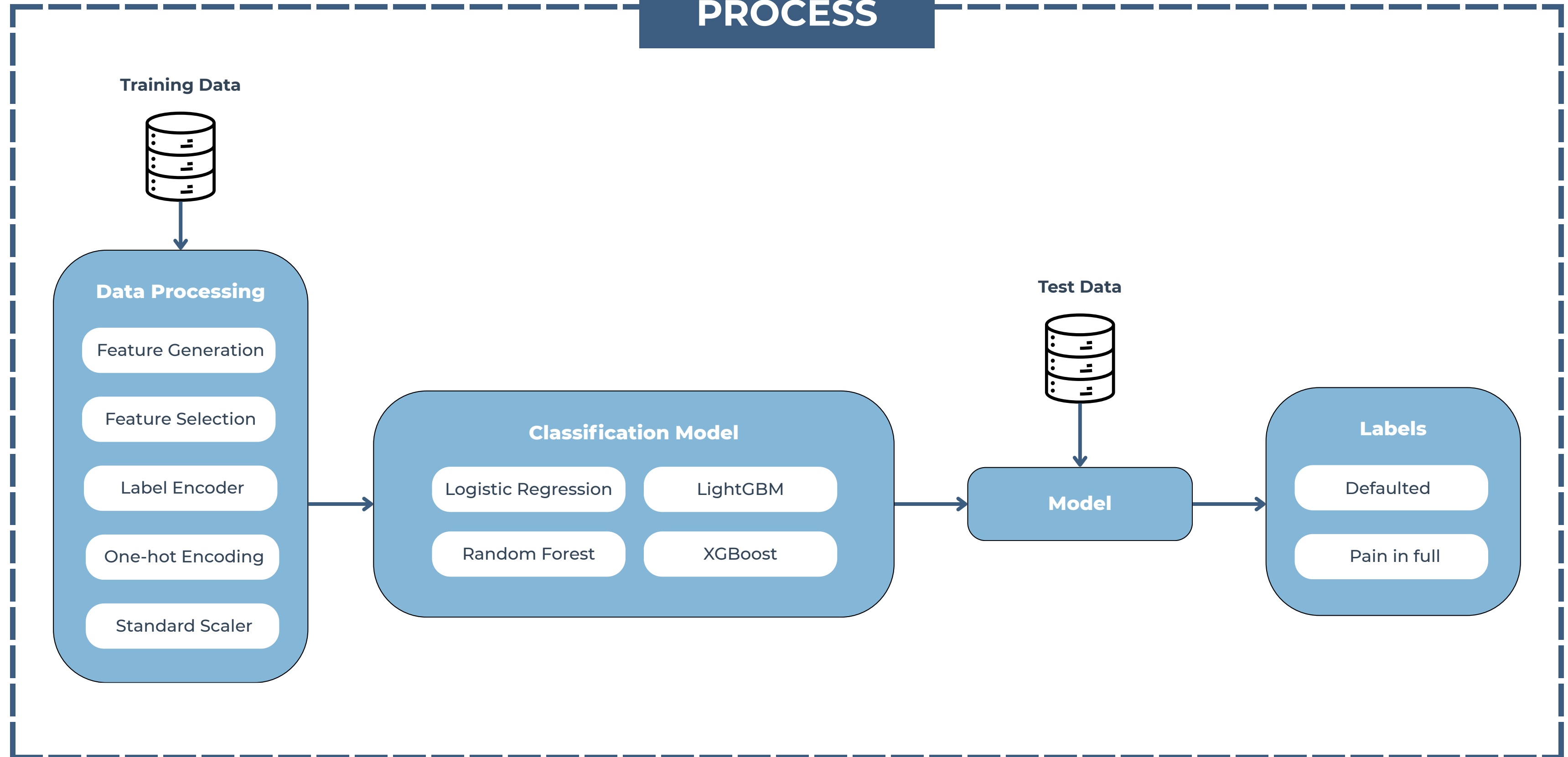


Figure 13. Customer default status by debt to saving ratio



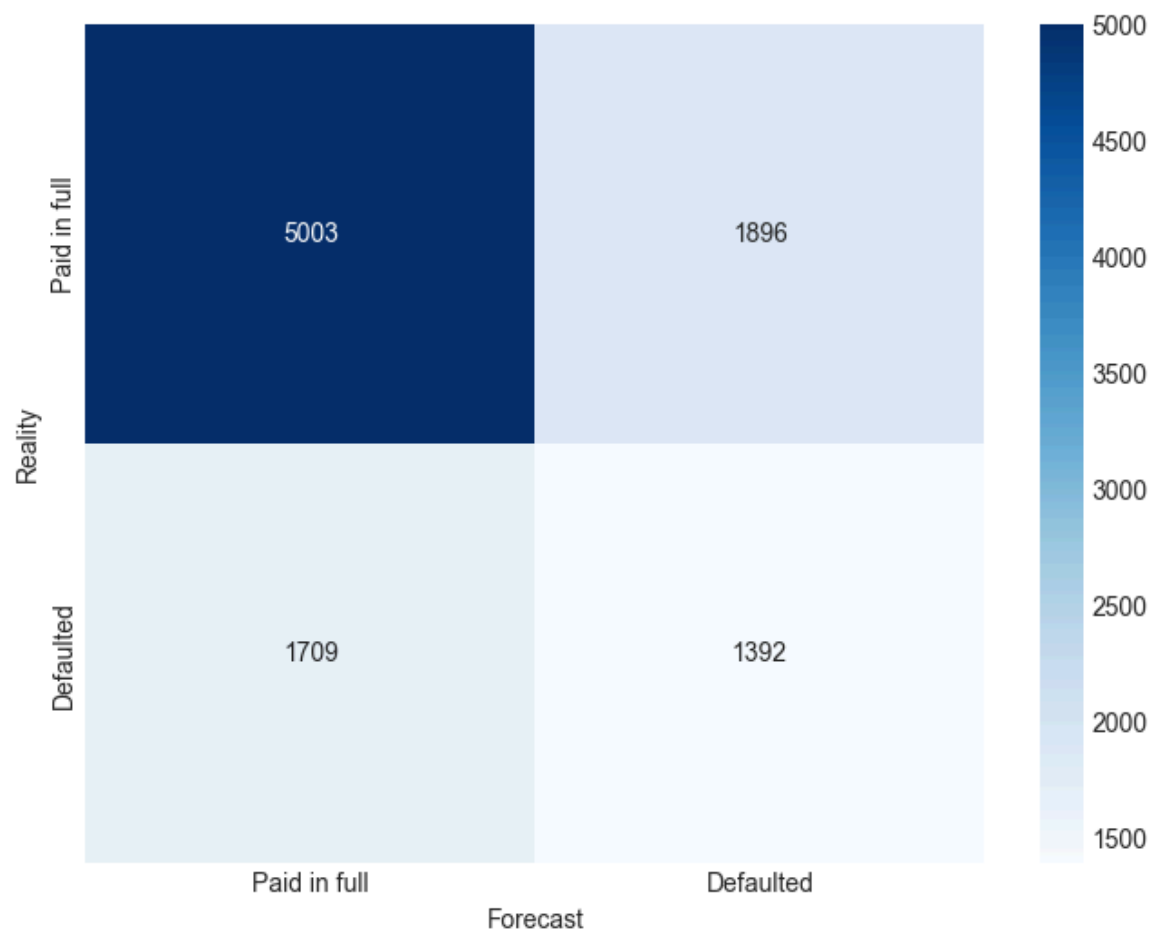
Borrowers with low income, poor credit, unstable jobs, and high debt-to-savings ratios are more likely to default, whereas higher-income, good-credit, full-time workers with stronger savings show significantly lower risk

PROCESS



LightGBM performance & Key drivers

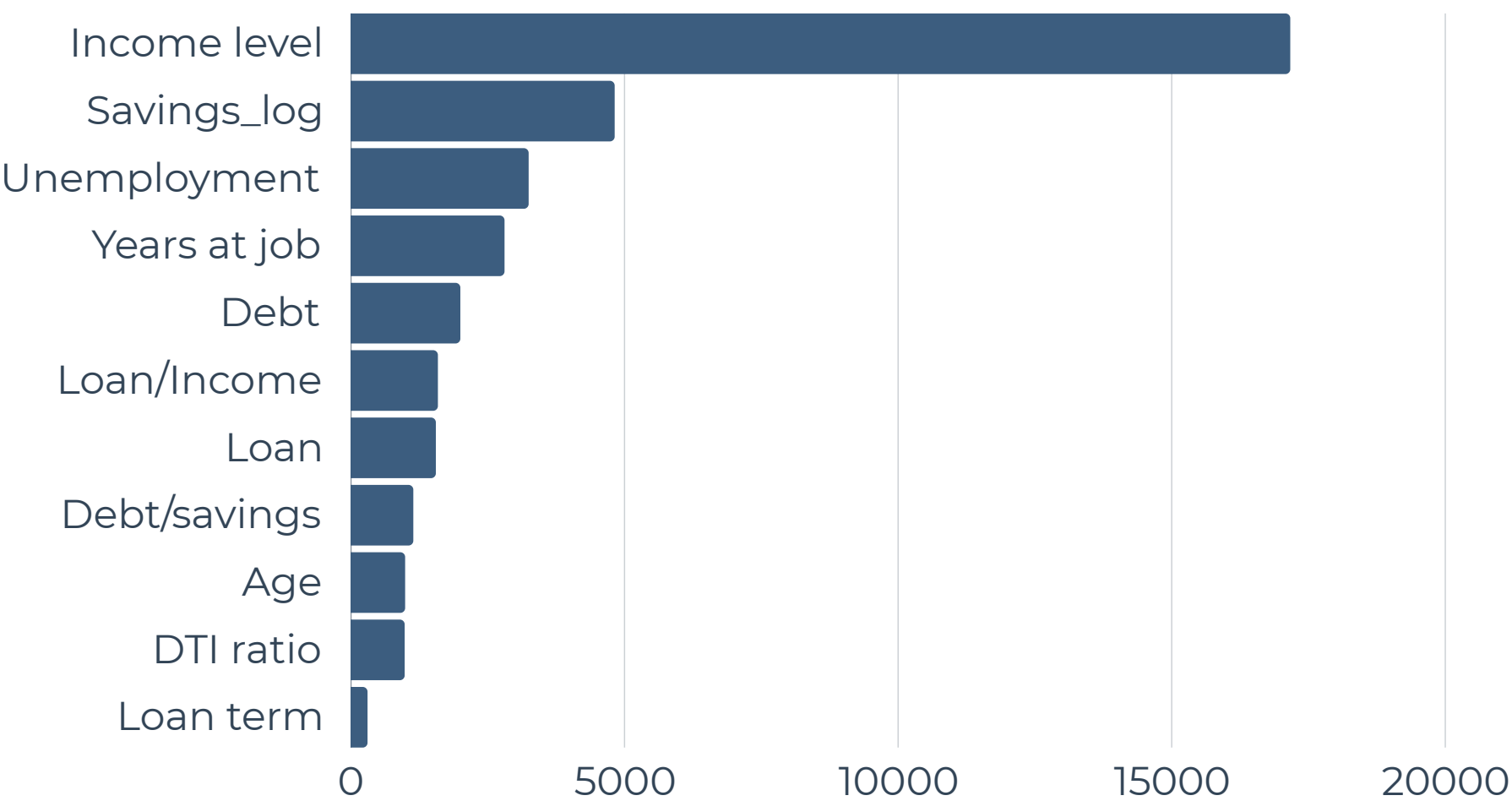
Figure 14. Confusion Matrix - LightGBM



Accuracy	Precision	Recall
0,6395	0,4230	0,4490

The 44.9% Recall shows that the model still **misses more than half of the problem customers**, a business risk that needs to be managed. The Precision-Recall trade-off also shows a trade-off: *to catch more risk, we may have to accept rejecting more good customers.*

Figure 15. Feature Importance



The Top 3 Factors: **Income Level**, **Savings**, and **Unemployment** are the most influential factors. This confirms that financial background and job stability are the “backbone” of credit assessment.

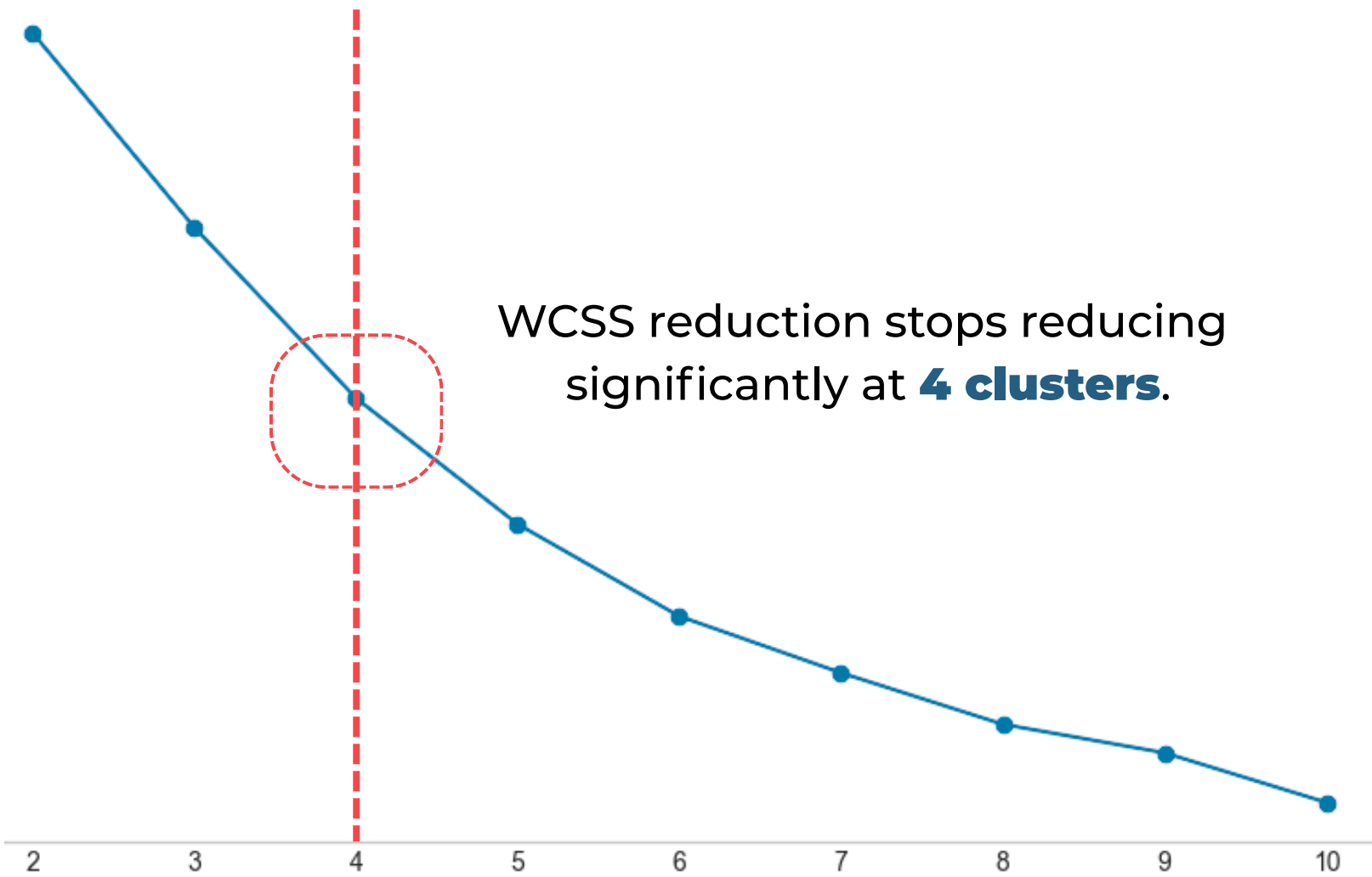
- ▶ Loan/Income
- ▶ Debt/Savings
- ▶ DTI ratio

Our created financial ratios are top predictors, demonstrating that the customer's financial context is more powerful for the model than absolute values alone.

Identifying Key Customer Archetypes through Data-Driven Segmentation

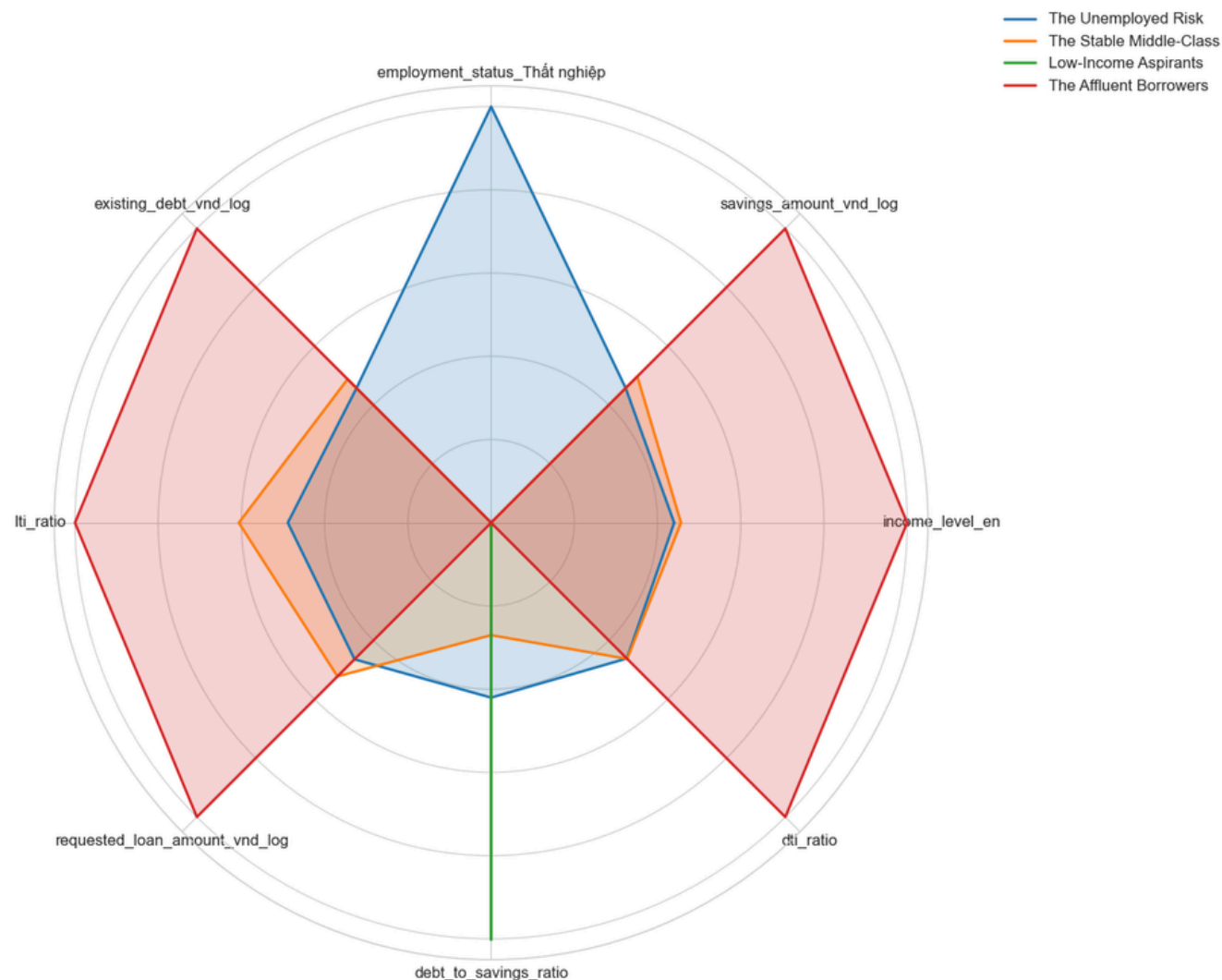
To identify distinct customer groups within our dataset, we employed the K-Means clustering algorithm.

Figure 16. Elbow Method



This point represents the best trade-off between capturing meaningful differences among groups and avoiding excessive complexity. Therefore, we proceeded with segmenting the customer base into four distinct archetypes.

Figure 17. Comparative Portrait of 4 Customer Segments



This segmentation moves us beyond a one-size-fits-all approach, enabling FinInnovate to develop targeted products, risk policies, and marketing strategies for each distinct customer archetype.

Segment	Size	Income level	Unemployment	Debt/Savings	Loan/Income	DTI ratio	Default rate
Segment 3: Wealthy, High Leverage	20.8%	2.2271	0.0	1.3125	37.8926	49.0415	17.8%
The wealthiest group with high income and savings; they take out large loans relative to their income but remain safest thanks to the lowest debt to savings ratio.							
Segment 1: Middle Class	44%	1.0610	0.0	2.6889	31.6171	39.9609	29.0%
This is the typical customer group, representing the majority of the borrowing population. They have stable employment and are in a decent financial position.							
Segment 2: Financially Strained	30.3%	0.0821	0.0	6.4083	21.9545	32.1432	40.5%
This is the group with the weakest financial resources (both income and savings are low). Their key defining characteristic is the severe imbalance between debt and savings.							
Segment 0: Unemployment Risk	4.9%	1.0257	1.0	3.4541	29.7406	39.9383	47.0%
This is clearly the highest-risk group due to the lack of a stable source of income. Although their income is not the lowest, their unemployment status is the decisive factor.							

Potential Customer Profile

1. Size:

15,133 (30.3%)
2. Savings amount:

52.4 triệu VND
3. Existing debt:

172.6 triệu VND
4. Requested loan amount:

122.4 triệu VND
5. DTI ratio:

32.14
6. Loan to Income ratio:

21.95
7. Debt to Savings ratio:

6.41
8. Income Distribution (Low):

91.8%
9. Income Distribution (Medium):

8.2%
10. Housing Distribution (Thuê nhà):

49.9%
11. Housing Distribution (Trả góp):

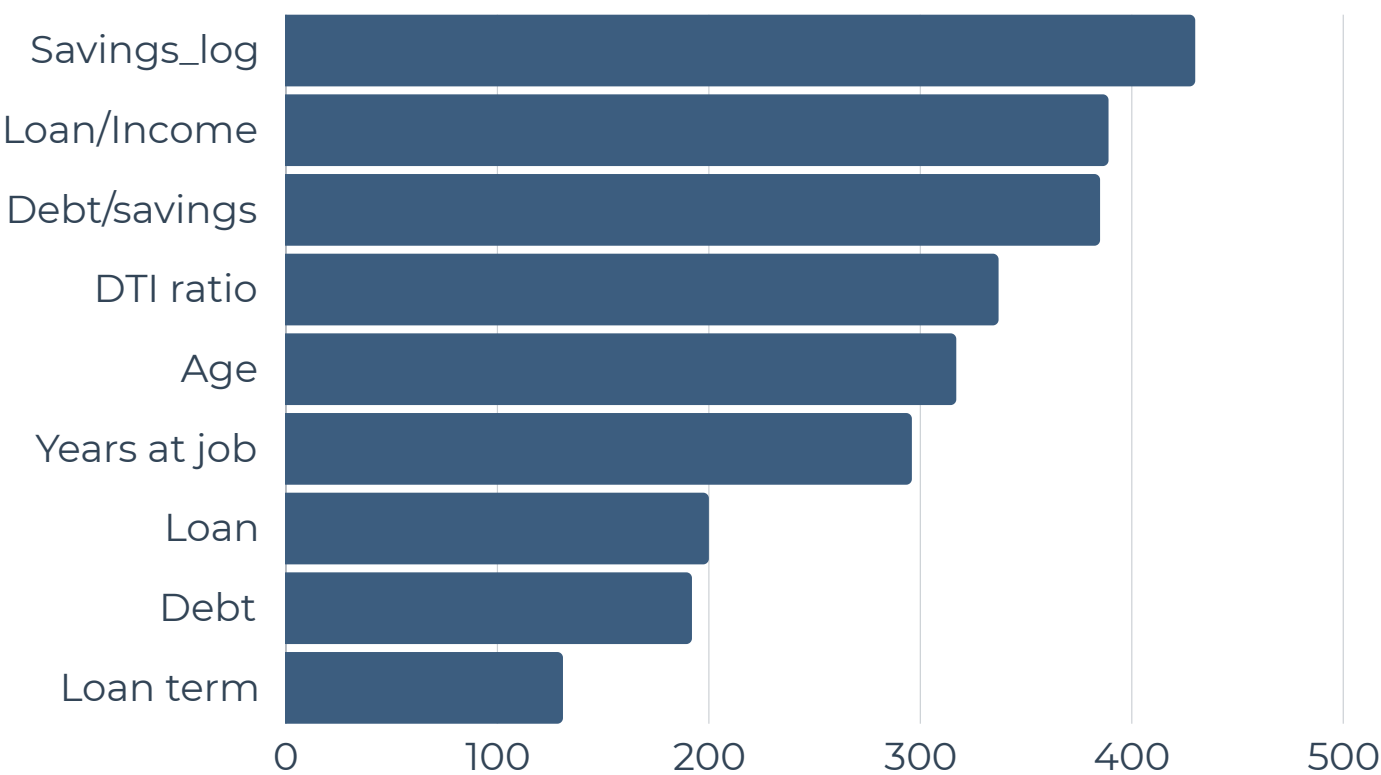
35.3%
12. Housing Distribution (Sở hữu):

14.9%
13. Default rate:

40.5%

We re-ran the LightGBM algorithm on the Segment 2 dataset.

Figure 1. Feature Importance - Segment 2



- ▶

The ability to save even a small amount is the strongest signal of financial discipline.
- ▶

How they manage their limited resources (e.g., Loan-to-Income ratio) is a better indicator of risk.
- ▶

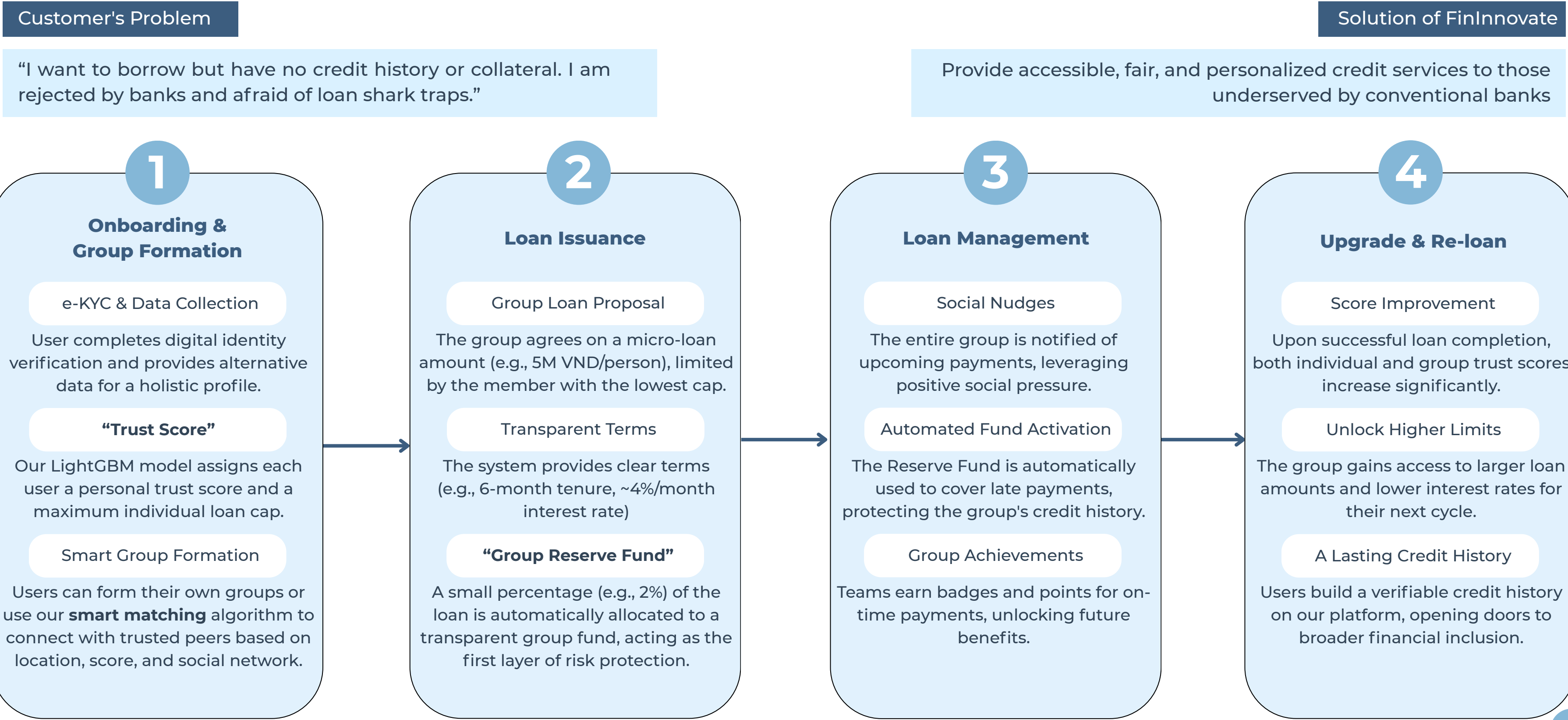
Stability factors like age and job tenure are crucial risk differentiators.
- ▶

How income is managed matters more than the amount itself.

Our Strategic Solution

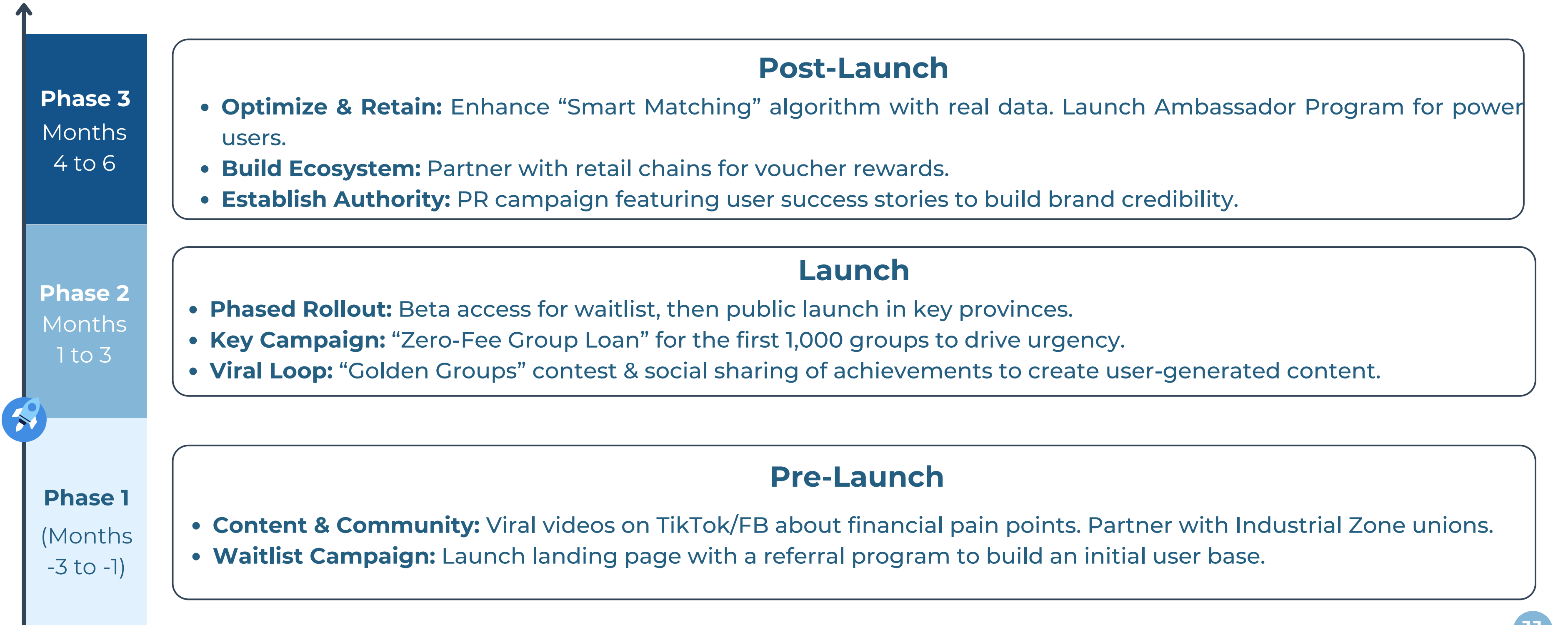
FinConnect

Smart Group Lending



Go-To-Market Strategy: Building a Trusted Community

Our 6-Month Goals: 10,000+ Registered Users | 500+ Active Loan Groups | #1 Trusted Brand for Underserved Workers



Market Research	Data Overview	Predictive Model	Segmentation	Recommendation
Roadmap				
Main actionable group	Q4/2025 (Pre-Launch)	Q1/2026 (Launch)	Q2/2026 (Activate)	Q3/2026 (Scale)
1. Product & Technology				
Core MVP Development (Scoring, Grouping, e-KYC)				
Beta & Public Launch				
UX/UI & Performance Optimization				
R&D for New Products (e.g., Group Savings)				
2. Go-To-Market & Growth				
Pre-Launch Marketing (Waitlist & Community Seeding)				
User Acquisition Campaigns (Digital Ads & Referrals)				
Brand Building & PR				
3. Community & Engagement				
Partnerships with Industrial Zones & Unions				
Launch Gamification & Ambassador Programs				
Ecosystem Integration (Retail Partners & Vouchers)				
4. Data Intelligence & Optimization				
Finalize & Deploy Trust Scoring Model v1				
Establish KPI Dashboards & Analytics				
Retrain & Optimize Models (v2) with Live Data				



FinConnect

Smart Group Lending

12

REFERENCE LIST

- 1.VTV. (2025) Thanh toán không tiền mặt tại Việt Nam tăng trưởng kép hàng năm, accessed 18 September 2025, <https://vtv.vn/thi-truong/thanh-toan-khong-tien-mat-tai-viet-nam-tang-truong-kep-hang-nam-20250618120134812.htm>
- 2.World Bank (2025) Cập nhật tình hình kinh tế Việt Nam: Bứt tốc phát triển nhân tài công nghệ cao của Việt Nam – Điểm lại Tháng 9/2025, accessed 18 September 2025, <https://i.ex-cdn.com/vietnamfinance.vn/files/attach/2025/09/10/bao-cao-tinh-hinh-kinh-te-vietnam-1214.pdf>
- 3.VnExpress (2025) GDP bình quân đầu người năm 2025 có thể đạt hơn 5.000 USD, accessed 18 September 2025, <https://vnexpress.net/gdp-binh-quan-dau-nguoi-nam-2025-co-the-dat-hon-5-000-usd-4848438.html>
- 4.VnExpress (2024) Thuê bao Internet di động lần đầu vượt 90% dân số, accessed 19 September 2025, <https://vnexpress.net/thue-bao-internet-di-dong-lan-dau-vuot-90-dan-so-4775151.html>
- 5.VietnamBiz (2023) Lượng người dùng smartphone tại Việt Nam ước đứng thứ hai Đông Nam Á vào năm 2026, thuộc top nhiều nhất thế giới, accessed 19 September 2025, <https://vietnambiz.vn/luong-nguoi-dung-smartphone-tai-viet-nam-uoc-dung-thu-hai-dong-nam-a-vao-nam-2026-thuoc-top-nhieu-nhat-the-gioi-202359104330221.htm>
- 6.VTV (2025) Cashless payments in Vietnam achieve double-digit annual growth, accessed 19 September 2025, <https://vtv.vn/thi-truong/thanh-toan-khong-tien-mat-tai-viet-nam-tang-truong-kep-hang-nam-20250618120134812.htm>
- 7.The Saigon Times (2025) Cashless payments exceed 12 billion transactions in half a year, accessed 19 September 2025, <https://thesaigontimes.vn/thanh-toan-khong-tien-mat-vuot-12-ti-giao-dich-trong-nua-nam/>
- 8.Báo Tin tức (2025) Ngân hàng tăng tốc số hóa, đẩy mạnh thanh toán không tiền mặt, accessed 20 September 2025, <https://baotintuc.vn/kinh-te/ngan-hang-tang-toc-so-hoa-day-manh-thanh-toan-khong-tien-mat-20250807191903467.htm>
- 9.Google Cloud (2025) Grounding with Vertex AI Search, accessed 20 September 2025, <https://cloud.google.com/vertex-ai/generative-ai/docs/grounding/grounding-with-vertex-ai-search>

APPENDIX

Suggested substitution variables for collection

New Variable Name	Description	Business Purpose & Rationale
marital_status	Categorical ('Single', 'Married', 'Divorced/Widowed')	Measures stability and responsibility. Married individuals often have more stable lives and greater financial responsibilities. The 'Divorced/Widowed' group may be experiencing a period of financial instability.
number_of_dependents	Integer (0, 1, 2, ...)	Measures financial burden. An individual with a 15 million income and 0 dependents has a vastly different repayment capacity than someone with the same income who must support two young children. The higher the number of dependents, the greater the risk.
education_level	Categorical ('High School', 'College/Vocational', 'University', 'Postgraduate')	A proxy for income potential and problem-solving ability. Individuals with higher education levels typically have higher future income potential and are more likely to find new employment quickly if they become unemployed.
time_at_current_address	Numeric (months/years)	Measures life stability. Similar to years_at_current_job, someone who moves frequently may have a less stable life than someone who has lived in one place for many years.
gender	Categorical ('Male', 'Female', 'Other')	In the consumer lending market, women often have lower access to credit than men, despite evidence suggesting they are better at repaying debts. Collecting this data helps the company understand if there are differences in borrowing behavior or default rates between gender groups, allowing for adjustments to marketing strategies or customer support accordingly.

New Variable Name	Description	Business Purpose & Rationale
linkedin_profile_completeness	Profile completion percentage (0-100%).	A complete profile (with a profile picture, summary, experience, skills) indicates professionalism and investment in one's career.
linkedin_connection_count	Number of connections (e.g., 0-500+).	A proxy for the size of one's professional network. A large network suggests the person is well-connected and may find it easier to seek opportunities or assistance.
device_model	Phone model name (e.g., 'iPhone 14 Pro', 'Samsung Galaxy A13').	A proxy for financial status. While not always accurate, owning a high-end device often correlates with better financial standing.
device_age	Time since the device was first activated.	A proxy for stability. Someone using the same phone for 2-3 years may have a more stable and less volatile lifestyle.
app_count_finance	Number of other financial apps installed (e-wallets, banking apps, investment apps).	Indicates the level of financial savviness. A person with many financial apps is typically more engaged and better at managing their money.
app_count_gambling	Number of gambling or betting apps installed.	An extremely strong risk signal. The presence of these apps is highly correlated with impulsive financial behavior and a higher probability of default.
contact_list_size	Number of contacts in the address book.	Measures the size of the user's social network.